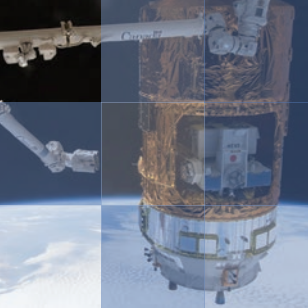




Angles in Standard Position

An angle in standard position is in the same quadrant as its terminal side.

Determining the Quadrant in Which an Angle Lies

Definitions

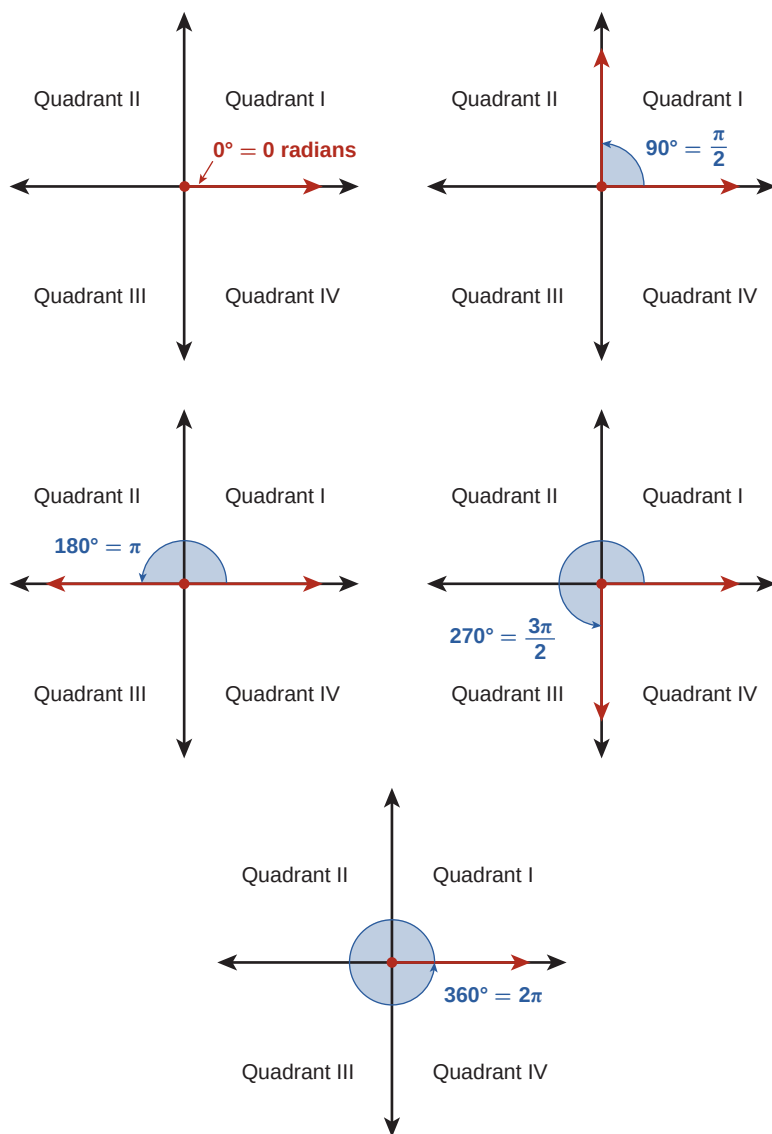
An angle is in **standard position** on the coordinate plane when its vertex is at the origin and its initial side is along the positive horizontal axis.

Quadrantal angles are angles in standard position that have their terminal sides along the horizontal or vertical axis.

► **Remember** The initial side of an angle is the ray from which the rotation of an angle starts. The terminal side of an angle is the ray at which the rotation of an angle stops.

Examples of quadrantal angles are 0° , 90° , 180° , 270° , and 360° . These degree measures are equivalent to radian measures of 0 , $\frac{\pi}{2}$, π , $\frac{3\pi}{2}$, and 2π , respectively.

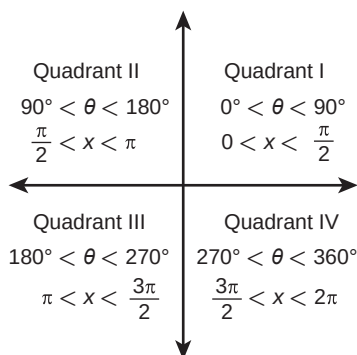
► **Remember** The first quadrant of the coordinate plane is between the positive horizontal axis and the positive vertical axis. The second, third, and fourth quadrants are located counterclockwise from the first quadrant, respectively.



An angle in standard position that is not quadrantal lies in the quadrant in which its terminal side lies.

Angles and Quadrants

The quadrant in which an angle θ lies can be determined by its measure.



EXAMPLE 1

Determine the quadrant in which the angle lies.

There are no units given for the angle measures in these examples, so they are understood to be in radians.

A $\frac{4\pi}{5}$

SOLUTION

Note that $\frac{4\pi}{5}$ is the same as $\frac{4}{5}$ of π . The fraction $\frac{4}{5}$ is between $\frac{1}{2}$ and 1.

So $\frac{4\pi}{5}$ is between the quadrantal angles $\frac{\pi}{2}$ (or $\frac{1}{2}$ of π) and π . The terminal

side of the angle is therefore in Quadrant II, so $\frac{4\pi}{5}$ lies in Quadrant II.

► **Think About It** Radian measures of angles given in terms of π can be thought of as fractional or integer multiples of π . To help visualize the standard position of such an angle, think about where it lies with respect to π .

For example, $\frac{7\pi}{6}$ is the same as $1\frac{1}{6}$ times π . It is located in the

third quadrant, $\frac{1}{6}$ of the way between π and 2π .

SOLUTION

Since the measure is not expressed in terms of π , convert it to degrees to determine the quadrantal angles between which it lies.

$$5.2 \left(\frac{180^\circ}{\pi} \right) \approx 298^\circ$$

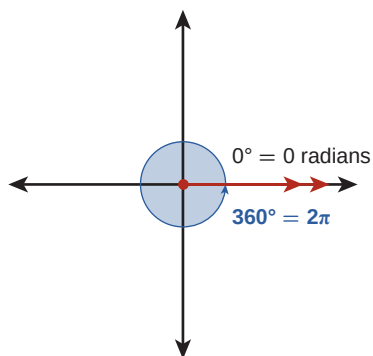
The measure 298° is between 270° and 360° , so the angle lies in Quadrant IV. ■

Generating and Identifying Coterminal Angles

Definition

Coterminal angles are angles in standard position that have the same terminal side but have different amounts of rotation. Their angles differ by a multiple of 360° , or 2π radians.

A simple example of coterminal angles is the quadrantal angles 0° (0 radians) and 360° (2π). Their measures differ by 360° , or 2π radians, but they both terminate on the positive horizontal axis.



How to Generate Coterminal Angles

Use the following expressions to determine the measures of angles that are coterminal with a given angle. Let n be an integer.

For angles in degrees, use $\theta + 360^\circ n$.

For angles in radians, use $x + 2\pi n$.

You can add or subtract multiples of 360° , or 2π , to determine the measures of angles coterminal with a given angle.

► Q&A

Q What is the degree measure of the angle between -360° and 0° that is coterminal with 90° ?

A -270°

EXAMPLE 2

Determine the measures of a positive angle and a negative angle that are coterminal with $\frac{13\pi}{4}$.

SOLUTION

The measure is in radians, so subtract multiples of 2π .

$$\frac{13\pi}{4} - 2\pi = \frac{13\pi}{4} - \frac{8\pi}{4} = \frac{5\pi}{4} \quad \text{positive angle}$$

$$\frac{13\pi}{4} - 2(2\pi) = \frac{13\pi}{4} - \frac{16\pi}{4} = -\frac{3\pi}{4} \quad \text{negative angle} \quad \blacksquare$$

EXAMPLE 3

Draw $-\frac{11\pi}{3}$ in standard position.

SOLUTION

Find an angle that is coterminal with $-\frac{11\pi}{3}$, but is between 0 and 2π .

Step 1 Add 2π as many times as you have to until you get to a value that is between 0 and 2π .

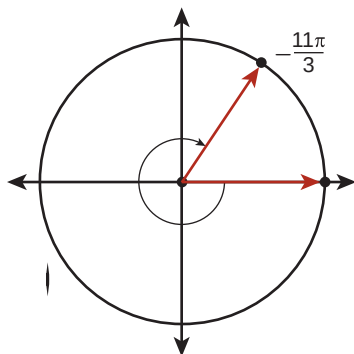
$$-\frac{11\pi}{3} + 2\pi = -\frac{11\pi}{3} + \frac{6\pi}{3} = -\frac{5\pi}{3}$$

$$-\frac{5\pi}{3} + 2\pi = -\frac{5\pi}{3} + \frac{6\pi}{3} = \frac{\pi}{3}$$

Step 2 Determine the quadrant in which the terminal side lies. The angle

$\frac{\pi}{3}$ is between quadrantal angles 0 and $\frac{\pi}{2}$, so it lies in Quadrant I.

Step 3 Draw the angle as coterminal with $\frac{\pi}{3}$. Label the angle $-\frac{11\pi}{3}$. ■

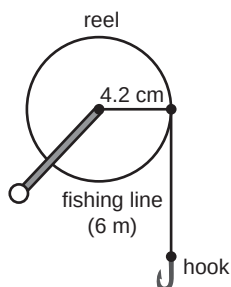


► **Think About It** Show rotation with an arrow pointing in the clockwise direction when drawing angles with negative measure.

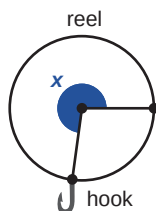
Applying Coterminal Angles

EXAMPLE 4

A fishing reel has 6 m of fishing line hanging from it, perpendicular to the positive horizontal axis that passes through the center of the reel, as shown in the first figure.



The fishing reel is rotated counterclockwise until the line is completely wrapped around the reel. The hook is now right up against the reel, as shown in the second figure.



What is the measure of angle x to the nearest hundredth of a radian?

SOLUTION

The reel rotates until the remaining 6 m of fishing line are wrapped around it, which generates an arc length of 6 m.

Step 1 Arc length and angle measure are proportional. Set up a proportion between the angle x and the arc length of 6 m.

$$\frac{x}{2\pi} = \frac{6}{2\pi(0.42)}$$

$$2\pi(0.42) \cdot \frac{x}{2\pi} = \frac{6}{2\pi(0.42)} \cdot 2\pi(0.42)$$

$$0.42x = 6$$

$$x \approx 142.8571$$

► **Think About It** To ensure better accuracy, round to the indicated place value after converting to the desired units.

► Q&A

Q How many full rotations does the fishing reel make?

A about 22

Step 2 The angle x is between 0 and 2π and is coterminal with 142.8571 radians. It is the difference of 142.8571 and a multiple of 2π . The quotient of 142.8571 and 2π is approximately 22.74. Subtract 22 multiples of 2π from 142.8571 to get $142.8571 - 22(2\pi) \approx 4.64$.

The measure of angle x is approximately 4.64 radians. ■